



اُنِيُوَرَسِيْتِي تِيَكْنُوْلُوْجِي مَارَا
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IMS560: ADVANCED DATABASE MANAGEMENT SYSTEM

**BALANCING CONSISTENCY AND SCALABILITY: WHY TRADITIONAL RDBMS REMAIN
RELEVANT IN MODERN WEB-SCALE APPLICATIONS**

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The raw facts, statistics, observations or information are known as data that can be processed and analyzed into valuable information. In the digital age, data is one of the most important resources of enterprises, as it helps in making decisions, running companies, and technical innovation. The digital technology is growing fast and this has transformed the manner various organizations gather, archive and process their data. Modern web applications like social media platforms, e-commerce websites, cloud applications, and Internet of Things (IoT) devices generate large quantities of structured and unstructured information on a daily basis. Information is often accessed in large and unstructured shapes or sizes, obtained through multiple different sources, and it is the task of the data analyst to initially observe, interpret, and develop an overall idea of the information before being able to extract it for further analysis (Magdum Rutuja, 2022). Database technology has evolved to meet the heightened demands of performance, scalability and availability as enterprises base their decision-making on data (Syed Abdul Rahman, 2023). RDBMS systems such as MySQL and PostgreSQL have been popular since defined schema, ensure high consistency, and provide reliability (Erickson, 2024). RDBMS can be scaled by optimisation strategies, but its scalability approach is often more difficult compared to distributed systems (Györödi et al., 2021).

With the emergence of Big Data, new NoSQL databases such as MongoDB and Apache Cassandra have been introduced and designed to support distributed computing and horizontal scaling. Cassandra, as an example, allows high availability and scalability in processing large size data on multiple servers. Consequently, there are experts who opine that traditional RDBMS are no longer suitable for existing web-scale systems due to scalability issues. This notwithstanding, RDBMS are not obsolete as they still can bring considerable advantages to data integrity, transactional reliability, security, and structured data management. The relational systems have also been scaled by modern advancements, such as advanced indexing, in-memory computations, and hybrid database structures, to ensure their applicability to the current data-driven world (Alshamrani et al., 2023). Consequently, although NoSQL databases offer advantages of scalability, traditional RDBMS are not irrelevant since they offer better data integrity, transactional reliability, security, and structured data management whilst adapting to the current scaling requirements.

The use of structured schemas is one of the greatest merits of traditional RDBMS. A fixed schema ensures that data will be stored uniformly within tables as well as connections which eliminates redundancy and enhances the accuracy of data. According to Wahyudi et al. (2022), database management systems like MySQL are still commonly used since their organized architecture allows for effective data organization and retrieval. Structured schemas come in handy where the accuracy of the data is required, including banking, health care, and

government. In addition, RDBMS offers intensive transaction processing with the ACID concepts, which guarantee atomicity, consistency, isolation, and durability. These attributes will ensure that the transactions are conducted properly even in case of a system failure. Alshamrani et al. (2023) remark, data security and integrity remains an important issue within the modern information systems, and reliable transactional services tend to become even more important. In spite of the emergence of NoSQL options, companies dealing with data of high sensitivity still heavily depend on relational databases. The continued use of RDBMS demonstrates that scaling is not the only factor when it comes to selecting a database. While NoSQL databases may handle bigger dispersed workloads, relational databases provide critical assurances on which many enterprises cannot compromise.

The issue of database security has emerged as a critical concern since companies are increasingly relying on online systems to store valuable information. Data breaches and illegal access may result in significant harm to finances and reputation. Alshamrani et al. (2023) report that contemporary information systems are facing growing security issues that require robust security measures. Relational databases have more sophisticated security measures such as user authentication, role-based access control, encryption, and audit trail. These characteristics enable businesses to track database usage and thwart illicit access. The force of law that requires stringent data access and modification in areas of health, finance, and government administration demands compliance with data protection regulations. Most companies are still using RDBMS because it has proven that it can maintain the security and integrity of data. This means that relational databases can be applied to a wider scope than scalability; they might also be applied to ensure important corporate information is secured.

Although they have their advantages, traditional RDBMS have challenges when attempting to execute web-based applications. Relational databases traditionally were constructed on vertical scalability, where more server resources such as processing capacity, memory, and storage are added. Although it is advantageous in certain situations, vertical scaling is progressively becoming more expensive and ultimately ceases to work due to hardware limitations. Khan et al. (2023) note that SQL databases are mainly tailored towards prepared structured data and vertical scalability, whereas NoSQL databases are tailored towards horizontal scalability and distributed architecture. With modern applications potentially generating massive volumes of data and supporting millions of users simultaneously, traditional database architecture might not be able to maintain high performance with enormous workloads. To test the social media example, billions of interactions are processed on social media sites daily. Handling this huge data volume on a single relational database server would result in performance issues and an increase in operating costs (Asif et al., 2024). These challenges gave rise to the creation of NoSQL databases that allow greater

scalability options. Though, just because scalability is constrained does not mean that it becomes obsolete. Rather, it concentrates on specific situations where other database systems might be more effective.

NoSQL databases were specifically created to be able to address the limitations of the traditional relational systems. Horizontal scalability is one of their significant advantages as it allows businesses to distribute data among a number of servers instead of relying on one computer. A popular example of a distributed NoSQL database is Apache Cassandra. The Apache Cassandra Documentation (n.d.) states that Cassandra is horizontally scaled, distributed and scalable in its schema management. Its architecture enables companies to scale by adding servers as workload increases, letting systems to scale without a major impairment in performance. Cassandra was created by Facebook to solve the issues of handling rapidly rising amounts of user-generated information and interactions. The distributed design of the system guarantees high availability and fault tolerance and it is suitable when it is used in large scale applications. Moreover, NoSQL databases are especially effective when it comes to semi-structured and unstructured data. NoSQL technologies are more flexible in terms of applications with constantly evolving data needs, as NoSQL supports flexible models, Syed Abdul Rahman (2023) affirms. This makes the systems more adaptable as compared to the traditional relational databases which require defined schemas. Consequently, Internet-based businesses such as Facebook and Netflix have adopted the NoSQL technology in order to provide services that are global in nature where availability and scalability are of significant importance.

NoSQL databases have better horizontal performance and scalability, but even scalability cannot provide information on whether a database system is obsolete or not. Most of these important systems, including financial systems, medical systems, and government databases, demand rigorous consistency, security, and transactional integrity. The defined schemas and the ACID-conformity of the transactions in relational databases often offer more support of these requirements. Although NoSQL databases are good at supporting large scale distributed workloads, they can compromise rigorous consistency in support of scalability and availability. Consequently, businesses are being forced to evaluate database systems based on their respective business requirements and not under the assumption that greater scalability will automatically make old RDBMS irrelevant. The online banking systems offer a practical example of the benefits of the dependability of the transactions. The transfer of money between accounts of one consumer to another has to be done properly and in a uniform way. Due to system failure, the cash can be debited in one account and no credit is provided to the account receiving it, leading to huge financial consequences. The ACID properties of relational databases help to prevent such cases as they provide a guarantee that transactions

will be either finished or not. Such a level of reliability is why financial companies still depend mostly on relational database technologies even though there are other database models.

Comparing MySQL and Cassandra, we can see why traditional RDBMS are not getting outdated. When it comes to customer relationship management systems, inventory management systems, and banking applications, MySQL is often employed due to its high level of consistency, structured schemas, and reliable processing of transactions (Erickson, 2024). Instead, Cassandra was meant to support distributed systems with high availability and horizontal scalability, it is suitable in large-scale applications where lots of data are handled by a large number of servers (Khan et al., 2023). Although Cassandra is good at handling distributed workloads, it might not be ideal with applications that require complex relationships between data items and strong transactional integrity. Salahuddin et al. (2024) found that SQL databases are still very successful for structured data management and transaction processing, however, NoSQL databases thrive in scalable and unstructured data settings. This analysis indicates that MySQL and Cassandra can satisfy different organization needs, and this means that the NoSQL databases can be used to complement, but not to substitute the traditional relational databases. This fact indicates that the problem lies not in a choice of a globally better database system. Rather, it focuses on the fact that various database systems are to fulfil different functional requirements. Consequently, the survival of RDBMS depends more on their applicability to particular tasks and less on their capability to compete with NoSQL databases on scalability only (Salahuddin et al., 2024).

Regardless of the development of NoSQL databases, relational databases continue to be extensively used in many industries. Financial institutions need relational databases that are dependable to handle transactions and ensure the balance in their accounts. Healthcare companies keep patient records, medical histories, and appointment systems that require high data accuracy and use RDBMS. In the same way, relational databases are used to store and access structured data by institutions and government entities. Erickson (2024) reports that MySQL is among the most popular database management systems in the world as it is reliable and highly supports transactional workloads. These illustrations show how most contemporary software opts to be consistent and durable as opposed to being as scalable as possible. Consequently, the fact that a number of industries are currently using relational databases constantly is a good indication that RDBMS continue to be relevant in the digital economy.

Whether it is due to their high scale or not, NoSQL databases have not supplanted the traditional relational databases. Rather, these two technologies co-exist since they fulfil different needs in an organization. According to recent studies, SQL and NoSQL databases are not mutually exclusive but complementary. Salahuddin et al. (2024) note that NoSQL

databases are extremely efficient with unstructured data and a horizontally scalable workload, but SQL databases are also very competent in managing structured data and performing transactions. Györödi et al. (2021) discovered that optimization techniques can enhance the speed of MySQL database and reduce scalability problems in relational database. In addition, recent RDBMS systems have superiorly grown on advanced technologies like clustering, replication, partitioning, and cloud-native implementation strategies. These innovations allow relational databases to go beyond their former limitations, and still retain integrity and reliability of the data. Cloud service providers have now many services that offer managed relational database with capacity to accommodate millions of transactions per day. Industry activities call into question the idea that RDBMS are outmoded. Such a large number of businesses still use relational databases as a transactional system in combination with NoSQL databases as analytics, caching, and content management at scale. The hybrid approach allows companies to take advantage of the strengths of both technologies.

The growing popularity of hybrid database designs goes to show that there is no inherent superiority in either RDBMS or in NoSQL databases. On the contrary, the current businesses are progressively combining the two technologies to enhance performance and reliability. Based on the latest studies on next-generation database systems, the hybrid architectures can support complex application requirements with SQL consistency and NoSQL scalability (Wajid et al., 2024). Such systems rely on relational databases to undertake mission critically transactions and on distributed NoSQL systems to process and analyse large volumes of data. MySQL can be used to handle client orders and payments, such as on an e-commerce site, whereas Cassandra can be utilized to store logs of user activity and suggestions. Having this segregation of duties ensures high consistency where it is needed and also enables high scaling of applications with large volumes. Consequently, instead of being substitutes to RDBMS, NoSQL databases have increased the variety of potential database systems. Companies are becoming more aware of the fact that effective data management would require the selection of appropriate technology to fit the task instead of merely relying on one database architecture.

Moreover, it appears questionable that either of the SQL or NoSQL will become the leader in the database administration in the future. Rather, business organizations are adopting the use of hybrid and multi-model database techniques that combine the two approaches. As Khan et al. (2023) highlight, database technologies have been shifting in line with varying needs of the corporations, which creates more adaptable and scalable database systems. Contemporary cloud services offer services which involve both relational and non-relational databases enabling the enterprises to choose the most suitable technology to do their business. Moreover, advances in distributed relational databases have eased much of

the scalability limitations, once linked with RDBMS. With the development of technology, the debate on which type of database is best replaces with the possibility of having several database types to work cooperatively. This trend suggests that RDBMS will remain to be a significant part of information systems to come, but will not be obsolete.

Finally, modern studies and industry experience do not confirm the notion that traditional RDBMS are no longer applicable to the present-day web-scale apps because of scale limits. Although NoSQL databases are associated with significant advantages of being horizontally scalable, having distributed processing, and capable of managing the schema so flexibly, conventional relational databases offer a higher level of data integrity, consistency, and transactional reliability. Recent studies indicate that new RDBMS systems have improved greatly due to optimization methods, integration with clouds and distributed design. Moreover, the growing adoption of hybrid database systems confirms that it is not mutually exclusive but complementary between relational and NoSQL databases. Lastly, successful companies know that one should not choose database based on technology dominance but rather the needs of the application. RDBMS will remain a significant part of the modern information system along with NoSQL, as the management of data is changing.

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